$$= 2I\left[\cancel{\int_0^d \bar{J} \times \bar{B}(2d)\,dy} + \cancel{\int_{2d}^{3d} i \times \bar{B}(x)\,dx} + \cancel{\int_0^d -\bar{J} \times \bar{B}(3d)}\right.$$

$$\left. + \cancel{\int_{2d}^{3d} -i \times \bar{B}(x)\,dx}\right]$$

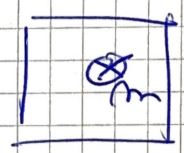
$$\bar{B}(2d) = \frac{\mu_0 I}{4\pi d}\,\hat{u}_z = \frac{\mu_0 I}{4\pi d}\,\hat{u}_z = \bar{B}(3d)$$

$$\Rightarrow \vec{F} = \vec{0}$$

$$i = -2I$$

e) $\tau = \vec{m} \times \vec{B}$ $\quad m = i S \vec{n}$

$$m = -2 I d^2 \vec{K}$$



$$\vec{B} = \frac{\mu_0 I}{2\pi}\left(\frac{3}{x} - \frac{1}{x-d}\right)\vec{K}$$

$$K \times K = 0 \qquad K \parallel K$$

$$\Rightarrow \boxed{\tau = 0}$$